

Derek Homan

SOFTWARE ENGINEER

Denver, CO

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Summary

Software engineer with 10+ years of experience building full stack applications, improving development processes, and solving complex technical problems. Known for intellectual curiosity and for asking thoughtful questions that challenge assumptions, uncover overlooked risks, and lead to better decisions. Brings that same mindset to AI assisted development, where clearly framing the problem and asking the right questions are often as important as the tools themselves. Enjoys learning; experimenting with new technologies; and turning ideas into practical, reliable systems.

Work Experience

ServiceTitan (Vertical SaaS company for Trades Companies)

Glendale, CA (remote)

SENIOR SOFTWARE ENGINEER

January 2022 - April 2025

- Helped develop and release a general ledger microservice, moving the product from beta toward general availability for customers
- Built a context-aware AI assistant with page-specific tools and schema-constrained responses, winning first place for “Most Impactful to ServiceTitan” in the May 2023 GenAI Hackathon
- Technologies: React, C#, .NET, SQL, generative AI

Gogo Inc. (inflight internet company)

Denver, CO

SENIOR DEVELOPER

April 2020 - January 2022

- Joined a team experiencing significant turnover and helped stabilize its development process and release a delayed project
- Introduced workflow improvements that improved coordination and helped move the project to release
- Migrated applications to AWS and modernized interfaces with responsive design
- Technologies: React, Redux, Backbone.Marionette, Python, CSS, AWS

Vendavo (pricing optimization software)

Denver, CO

SENIOR .NET DEVELOPER

June 2018 - April 2020

- Ported and Redesigned an existing Silverlight project into React
- Helped coworkers adopt Git after the team migrated from TFS by creating and delivering a presentation, then supporting rebases and repository issues over the following months
- Created proof-of-concept hybrid mobile application that worked offline and synchronized orders when connectivity returned
- Technologies: React, Redux, C#, .NET, Ionic

CSG International (ordering, and CRM products for ISPs)

Omaha, NE

SOFTWARE DEVELOPER ENGINEER SENIOR

July 2016 - June 2018

- Implemented unit and automated integration tests
- Provided estimates and implemented new features
- Developed and tested bugfixes and back-ported them in Hotfixes and Patches
- Presented technical talks to coworkers, and provided feedback via Code Reviews
- Technologies: Angular, Redux

idea5 (startup that aggregates data and generates reports for financial institutions)

Omaha, NE

FULL STACK SOFTWARE DEVELOPER

December 2013 - July 2016

- Improved slowest page load from 15-30 seconds to 2-3 seconds (by proposing and implementing a microservice) to generate thumbnails instead of live rendering
- Discovered a security flaw of a t4 template exposing sensitive application details such as connection strings
- Designed and implemented microservices using Docker, Node.js, Elasticsearch, and Windows Azure
- Wrote the payment checkout system utilizing C# and Authorize.Net
- Mentored an intern and presented several talks to fellow coworkers
- Technologies: Backbone Marionette, C#, .NET, Azure, Docker, Node.js

Nebraska Book Company (online web commerce bookstore)

Lincoln, NE

.NET DEVELOPER

June 2012 - December 2013

- Wrote, maintained and tested backend services used to manage orders and inventory
- Gathered requirements and developed specifications for projects involving multiple departments
- Maintained, updated, and documented several legacy programs inherited from other programmers
- Technologies: C#, VB ASP.NET, ASP.NET

UCARE, University of Nebraska - Lincoln (embedded devices research group)

Lincoln, NE

RESEARCH ASSISTANT

May 2010 - May 2012

- Developed tracking devices (backpacks with sensors and GSMs) for Sandhill cranes as an undergraduate research assistant for Dr. Vuran
- Performed field testing and gathered data
- Wrote drivers for various embedded devices
- Assembled and tested PCB sensor boards
- Technologies: C++, TinyOS

Interact Incorporated (software development for ISPs and call centers)

Lincoln, NE

INTERN

May 2009 - May 2010

- Found and fixed a critical issue blocking a project from release caused by bug in third party software (reverted the dependency)
- Designed and implemented flows for automated customer service portals using VXML (Voice XML)
- Worked on a project to automate generation of VXML applications with a GUI
- Technologies: bash, VXML, Java, C++

Phoenix Web Group (web design and hosting company)

Waverly, NE

DEVELOPER

January 2006 - December 2008

- Front end design using Photoshop, CSS, and HTML
- Wrote scripts to automate testing web applications
- Developed, designed, and tested web applications
- Made an internal service to generate invoices for customers
- Technologies: C# ASP.NET, HTML, CSS

Skills

Programming

C, C++, C#, Java, LaTeX, Javascript

Web

HTML, CSS, LESS, SASS, Angular, Backbone Marionette, React, Redux, Express

Platforms / Task Runners

Azure, AWS, Docker, Serverless Framework, Ansible, Node.js, gulp.js

Education

University of Nebraska - Lincoln

Lincoln, NE

B.S. IN COMPUTER ENGINEERING

May 2012

Honors & Awards & Achievements

- 2023 **First Place — GenAI Hackathon**, ServiceTitan, Category: “Most Impactful to ServiceTitan Hack”
- 2007 **Regents Scholarship**, University of Nebraska - Lincoln
- 2007 **Eagle Scout**, Boy Scouts of America

Presentations

Nebraska Code

Lincoln, NE

PRESENTER FOR <A NEWB'S GUIDE TO GETTING STARTED WITH IONIC>

2015

- Guide on what the Ionic framework (hybird mobile app framework with Angular and Cordova) is and how to make a simple application with it

Extracurricular Activity

Omaha Coder Dojo

Omaha, NE

MENTOR

2014

- Helped 8-12 year olds learn how to code (advice, explanations, troubleshooting, etc)

Side Projects

CURRENT PROJECTS

2025 - Present

- Contributed to openclaw: found its clawdock script failing on custom ports, reported the bug, and opened a fix (PR 75754, merged via PR 93663 after going stale)
- Built tooling and feedback loops that help coding agents inspect results, diagnose failures, retry safely, and surface verification status, project context, and token cost

base-stack

APPLICATION TEMPLATE & AGENTIC CI

2025 - Present

- Application Template that synchronizes shared code into downstream projects (Stonks, Eunomia) through GitHub Actions: a push to **main** opens a pull request in each downstream repo, with per-file merge strategies and exclusions protecting application-specific code
- Built a reverse channel: where an **/upstream <instructions>** comment on a downstream pull request opens an issue on the template with the source diff attached, where a Claude agent implements the fix so it re-propagates out to every downstream project
- Reference implementation is an authenticated CRUD app where the browser holds a reactive Convex subscription authenticated by a Clerk JWT, and every Convex function derives the caller from the auth context rather than a client-supplied user id, re-checking ownership on each access
- Technologies: Next.js 15, React 19, Clerk, Convex, Tailwind / shadcn, GitHub Actions, vitest; ships as a non-root Docker image to a self-hosted Coolify VPS that runs 3-4 apps, all downstream of this template

Stonks

OPTIONS FLOW ANALYSIS & AGENTIC STRATEGY BACKTESTING

2025 - Present

- Options flow platform where an LLM agent proposes trading strategies and a sandboxed Python harness backtests them. The agent never writes to the database directly; every proposal lands in a human approval inbox, and approval is what creates the strategy and queues the backtest
- Watermark-based incremental ingestion of Unusual Whales options flow on a 5-minute schedule, feeding a sliding-window detector for repeated-strike, premium-spike, and sweep-cluster patterns (deduped so an ongoing cluster fires once, not every cycle)
- User-defined alerts over live flow with per-alert watermarks, cooldowns, and daily firing caps; alerts can be gated by an LLM triage step that classifies a matched batch as noteworthy or routine noise before notifying (email / Slack / Discord)
- Backtest harness runs agent-authored Python strategies through an AST guard and isolated subprocess with a hard timeout, then feeds failures into a bounded diagnose-fix-verify loop limited to agent-originated code
- Technologies: Next.js 16, React 19, Convex, Clerk, Trigger.dev, Python (backtesting.py, pandas-ta, yfinance), Anthropic SDK, Tailwind / shadcn, vitest

Eunomia

MULTI-AGENT APP FOR DISCOVERING, EVALUATING, AND EXECUTING BUSINESS IDEAS

2025 - Present

- Built a durable LangGraph orchestration state machine on Convex that plans specialist work, requests human approval, dispatches parallel research waves, joins asynchronous results, and resumes from persisted checkpoints
- Made execution replay-safe through atomically idempotent wave dispatch and a deterministic completion coordinator, preventing duplicate specialist jobs or continuation turns during retries and concurrent completions
- Added an evidence-quality pipeline that checks report provenance and uses a structured-output critic to block unsupported research before the orchestrator synthesizes findings or proposes the next confirmed wave
- Six specialist agents perform web-grounded market, financial, legal, planning, go-to-market, and evaluation work; outputs are stored as versioned artifacts and scorecards and streamed back into the originating conversation
- Technologies: Next.js, React, Convex, LangGraph, Clerk, Anthropic SDK, Perplexity, Tailwind / shadcn, Vitest, Docker / Coolify

Dollar Game Picks (dollargamepicks.com)

CO-FOUNDER

2015-2016

- Failed startup where we provide predictions on whether or not to bet on the line for NFL games for a dollar and if we are wrong we give you a credit for another prediction
- Wrote the website, Stripe integration, account management / registration, email automation (Sendgrid)
- Technologies: Sequelize, Node.js, Angular, Express

Frigg.io

FOUNDER

2018-2020

- Wedding planning website using text messages to provide info and updates to attendees
- Technologies: Angular, Firebase, Twilio

Hobbies/Miscellaneous

Hobbies

hiking, reading, cooking, working on side projects, weightlifting, meditation

Favorite Books

Lean Startup, A Philosophy of Software Design, The Design of Design, Getting Things Done, The Courage to be Disliked, Let My People Go Surfing, An Astronaut's Guide to Life on Earth

Personal Preferences

Git Rebases » Merges, Vim, Visual Studio Code, tmux, openclaw, hermes-agent, 2 space tabs (although project consistency trumps preference)